Natural Transition Orbitals (NTOs)

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This section explains how excitation amplitudes form a transition matrix and how its singular value decomposition yields natural transition orbitals. It retains the same transition information, while a small number of hole-electron pairs may provide a compact interpretation.

1 Central Idea

In an electronic excitation, the transition matrix **T** organizes, in a single array, the amplitudes connecting occupied orbitals to virtual orbitals. The index i labels an occupied orbital, and the index a labels a virtual (unoccupied) orbital. Each matrix element answers the question: What is the amplitude of the excitation from the occupied orbital i to the virtual orbital a?

$$T_{ia} = \ amplitude\ of\ excitation\ i \rightarrow a.$$

By convention, rows represent occupied orbitals and columns represent virtual orbitals. Thus, if there are n relevant occupied orbitals and m relevant virtual orbitals, **T** has dimensions $n \times m$. Throughout, orbital bases are assumed to be orthonormal. The finite dimensions count the orbitals actually retained, not necessarily every orbital in a calculation. A matrix entry is generally a dimensionless, potentially complex amplitude. The arrow indicates the physical excitation assignment, not yet the direction in which a one-electron matrix acts on a vector.

In a description based on spin orbitals, a single excitation $i \rightarrow a$ can be represented by an excited Slater determinant obtained by replacing an occupied spin orbital i with a virtual spin orbital a in the reference determinant. A Slater determinant is an antisymmetric many-electron wave function constructed from spin orbitals, each with a spatial part and a spin part.

$$\Phi_i^a = \hat{a}_a^\dagger \hat{a}_i \Phi_0 \cdot$$

Here, Φ_0 is the reference determinant. The annihilation operator $\hat{a}_i$ removes an electron from spin orbital i, and the creation operator $\hat{a}_a^\dagger$ places an electron in a virtual spin orbital a. The dagger denotes the Hermitian adjoint. Thus, Φ_i^a is a singly excited Slater determinant, with its phase fixed by the stated operator ordering. However, there is an important distinction. When spatial orbitals and states with well-defined total spin are used, a "configuration" such as HOMO → LUMO may correspond to a spin-adapted combination of two determinants rather than to one determinant. Such a combination is called a configuration state function (CSF). HOMO means highest occupied molecular orbital; LUMO means lowest unoccupied molecular orbital. For a closed-shell singlet reference, with total-spin quantum number $S = 0$ and spin functions α and β, the singlet CSF is

$$\mathrm{CSF}_{i\rightarrow a}^{S=0} = \frac{1}{\sqrt{2}}\left(\Phi_{i_\alpha}^{a_\alpha} + \Phi_{i_\beta}^{a_\beta}\right)$$

For a singlet state, the appropriate combination of the α- and β-excited determinants produces a spin-adapted function. In configuration interaction singles (CIS), the Tamm–Dancoff approximation (TDA), and time-dependent density functional theory (TD-DFT), "configurations $i \rightarrow a$" are often discussed informally, although the implementation may use spin-adapted combinations. The precise statement is therefore: Each single excitation in a spin-orbital representation corresponds to a Slater determinant; a spin-adapted spatial configuration may correspond to a CSF containing more than one determinant.

The plus sign in the singlet CSF assumes that both excited determinants are defined by the creation-annihilation operator convention shown above. For orthonormal spin orbitals, the two determinants are orthogonal and individually normalized. Writing them as A and B, their normalized sum satisfies

$$\langle \frac{A+B}{\sqrt{2}} | \frac{A+B}{\sqrt{2}} \rangle = \frac{1}{2}(\langle A|A\rangle + \langle A|B\rangle + \langle B|A\rangle + \langle B|B\rangle) = 1$$

For HOMO → LUMO, this CSF combines the α-spin promotion and the β-spin promotion with equal coefficients $1/\sqrt{2}$. Reordering spin orbitals inside a determinant can change its written sign; a spin-adapted combination must use a consistent determinant-phase convention.

2 Constructing the T Matrix

Consider two relevant occupied orbitals, HOMO−1 and HOMO, and two virtual orbitals, LUMO and LUMO+1. Here, HOMO−1 is the occupied orbital immediately below the HOMO, and LUMO+1 is the virtual orbital immediately above the LUMO in the chosen orbital-energy ordering. Let H and L abbreviate HOMO and LUMO in determinant labels. Suppose the excited state is represented schematically by the following expansion, denoted Ψ^*. In this state label only, the superscript asterisk denotes an excited state, not complex conjugation.

$$\Psi^* = -0.40\,\Phi_{H-1}^{L} + 0.20\,\Phi_{H-1}^{L+1} + 0.60\,\Phi_{H}^{L} + 0.30\,\Phi_{H}^{L+1}$$

Place the amplitudes in **T**, retaining their signs (relative phases). Do not square the coefficients.

occupied / virtual	LUMO	LUMO+1
HOMO−1	−0.40	0.20
HOMO	0.60	0.30

$$\mathbf{T} = \begin{pmatrix} -0.40 & 0.20 \\ 0.60 & 0.30 \end{pmatrix}$$

For example, if a program output lists 44 → 46 = −0.400, 44 → 47 = 0.200, 45 → 46 = 0.600, and 45 → 47 = 0.30, identify 44 = HOMO−1, 45 = HOMO, 46 = LUMO, and 47 = LUMO+1, and fill the corresponding matrix entries.

The four displayed coefficients are retained exactly. For orthonormal excited determinants, their squared norm is $0.40^2 + 0.20^2 + 0.60^2 + 0.30^2 = 0.65$, not 1. Thus, the displayed state is a schematic, unnormalized expansion or a retained part of a larger expansion. If these are the complete coefficients of a state, divide each by $\sqrt{0.65}$ to normalize it. This common scaling changes the singular values but not the NTO directions.

3 A Second Example

Now consider three relevant occupied orbitals, HOMO−2, HOMO−1, and HOMO, and three virtual orbitals, LUMO, LUMO+1, and LUMO+2. The offsets −2 and +2 indicate positions two

below the HOMO and two above the LUMO, respectively. Suppose the excitation amplitudes are as follows:

HOMO−2 → LUMO: 0.10
HOMO−2 → LUMO+1: −0.20
HOMO−2 → LUMO+2: 0.05
HOMO−1 → LUMO: 0.35
HOMO−1 → LUMO+1: 0.40
HOMO−1 → LUMO+2: −0.15
HOMO → LUMO: 0.55
HOMO → LUMO+1: −0.25
HOMO → LUMO+2: 0.30

The excited state can be written schematically as

$$\Psi^* = 0.10\Phi_{H-2}^{L} - 0.20\Phi_{H-2}^{L+1} + 0.05\Phi_{H-2}^{L+2} + 0.35\Phi_{H-1}^{L} + 0.40\Phi_{H-1}^{L+1} - 0.15\Phi_{H-1}^{L+2} + 0.55\Phi_{H}^{L} - 0.25\Phi_{H}^{L+1} + 0.30\Phi_{H}^{L+2}$$

The rows follow the selected order of occupied orbitals, and the columns follow the selected order of virtual orbitals:

occupied / virtual	LUMO	LUMO+1	LUMO+2
HOMO−2	0.10	−0.20	0.05
HOMO−1	0.35	0.40	−0.15
HOMO	0.55	−0.25	0.30

$$\mathbf{T} = \begin{pmatrix} 0.10 & -0.20 & 0.05 \\ 0.35 & 0.40 & -0.15 \\ 0.55 & -0.25 & 0.30 \end{pmatrix}$$

For example, the entry in row 2, column 3 is $T_{23} = -0.15$, because it represents the amplitude for HOMO−1 → LUMO+2. The entry in row 3, column 1 is $T_{31} = 0.55$, because it represents HOMO → LUMO.

The nine entries have a squared norm 0.8125. If the expansion is complete and the configurations are orthonormal, normalization requires division by $\sqrt{0.8125}$. If it is a selected block from a larger calculation, keep its actual amplitudes when representing that block. Never renormalize an incomplete block without stating that choice.

4 What Happens When an Excitation Is Not Listed?

If a particular excitation $i \to a$ makes no significant contribution, its matrix entry may be treated as approximately zero:

$$T_{ia} \approx 0.$$

Thus, a $\mathbf{T}$ matrix may contain many zeros or very small entries. In actual calculations, programs often print only amplitudes above a threshold; omitted terms are not necessarily exactly zero. If a program prints only entries with $|T_{ia}| \geq 0.05$, an unprinted amplitude of 0.03 may be approximated by zero, but it is not known to vanish. Its omission contributes $0.03^2 = 0.0009$ to the sum of squared absolute errors in all matrix entries.

The matrix $\mathbf{T}$ contains amplitudes, not probabilities. The relative signs of the coefficients affect the linear combinations that form the NTOs. Let X_{ia} denote an excitation amplitude in the example model or TDA convention under consideration. Then

$$T_{ia} = X_{ia} \ \ (example\ model/TDA), \quad T_{ia} \neq |X_{ia}|^2.$$

Squared magnitudes may subsequently enter weights or contributions, but they must not replace amplitudes when assembling the transition matrix.

Consider the real matrix $\mathbf{A}$ below, used only to illustrate phase information. Its two rows are orthogonal, so both singular values are equal to 1. Squaring its entries gives a different matrix $\mathbf{B}$, which has rank one and singular values 1 and 0. Here, rank is the number of linearly independent columns.

$$\mathbf{A} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{A}\mathbf{A}^{\dagger} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Thus, squaring the amplitudes changes both the rank and the NTO structure. For a complex coefficient, a weight uses $|X_{ia}|^2 = X_{ia}^* X_{ia}$, not the generally complex quantity X_{ia}^2.

5 From the T Matrix to NTOs: SVD

After constructing $\mathbf{T}$, apply the *singular value decomposition* (SVD). In the full SVD, $\mathbf{U}$ is an $n \times n$ unitary matrix, $\mathbf{V}$ is an $m \times m$ unitary matrix, and $\mathbf{\Lambda}$ is an $n \times m$ rectangular diagonal matrix whose diagonal entries are nonnegative singular values. The dagger denotes conjugate transpose, so $\mathbf{V}^{\dagger} = (\mathbf{V}^*)^{\mathbf{T}}$, where the asterisk means entrywise complex conjugation and the superscript T means transpose. For real matrices, unitary matrices are orthogonal, and the dagger reduces to the transpose.

$$\mathbf{T} = \mathbf{U}\mathbf{\Lambda}\mathbf{V}^{\dagger}$$

The matrix $\mathbf{U}$ provides linear combinations of occupied orbitals, while $\mathbf{V}$ provides linear combinations of virtual orbitals. The singular values occupy the diagonal of $\mathbf{\Lambda}$. Notation: Let $q =$

$min(n, m)$, and order the singular values as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_q \geq 0$. Let r_T denote the rank of **T**, the number of nonzero singular values. Sums over transition pairs run from 1 to q, or equivalently over the r_T nonzero terms. Unpaired null-space columns of a full rectangular SVD carry no transition amplitude. A reduced SVD retains only the needed columns.

Let ϕ_i^{occ} and ϕ_a^{virt} denote the original orthonormal occupied and virtual molecular orbitals; the superscripts "occ" and "virt" are labels, not powers. The index k labels an NTO pair, and the superscripts h and e denote its hole and electron members. The matrix elements U_{ik} and V_{ak} are their expansion coefficients. The defining equations are

$$\psi_k^h = \sum_i U_{ik}\, \phi_i^{\mathrm{occ}}$$

and

$$\psi_k^e = \sum_a V_{ak}\, \phi_a^{\mathrm{virt}}.$$

In the first equation, ψ_k^h is the kth hole NTO. It describes the orbital character associated with removing electronic amplitude during the excitation. The index k identifies the NTO pair: $k = 1$ for the first pair, $k = 2$ for the second pair, and so on. The occupied-orbital index i runs over all $N_{occ} = n$ occupied orbitals retained in the matrix:

$$i = 1,2,3, \ldots, N_{\mathrm{occ}}.$$

The term

$$\phi_i^{\mathrm{occ}}$$

is the ith original occupied molecular orbital, for example,

$$HOMO, \quad HOMO - 1, \quad HOMO - 2, \ldots$$

The coefficient U_{ik} specifies the contribution of the occupied orbital i to hole NTO k. For example, the source illustrates the following linear combination:

$$\psi_1^h = 0.80\, \phi_{HOMO} + 0.50\, \phi_{HOMO-1} - 0.20\, \phi_{HOMO-2}\,.$$

Normalization of illustrative NTO coefficients. The source's hole coefficients (0.80, 0.50, −0.20) have squared norm 0.93. The expression above is therefore an unnormalized illustrative combination, not yet a normalized NTO. The corresponding normalized hole orbital is

$$\psi_1^h = \frac{0.80\phi_{HOMO} + 0.50\phi_{HOMO-1} - 0.20\phi_{HOMO-2}}{\sqrt{0.93}}$$

This expression means that the first hole orbital is a linear combination of those three occupied orbitals; it need not equal the HOMO. The second defining equation,

$$\psi_k^e = \sum_a V_{ak}\, \phi_a^{\text{virt}},$$

is entirely analogous, but it involves virtual orbitals. Here,

$$\psi_k^e$$

is the kth electron, or particle, NTO. It describes the orbital character associated with adding electronic amplitude during the excitation. The virtual-orbital index a runs over the $N_{\text{virt}} = m$ retained virtual orbitals:

$$a = 1,2,3, \ldots, N_{\text{virt}}\,.$$

Examples include LUMO, LUMO+1, and LUMO+2. The function ϕ_a^{virt} is an original virtual orbital, whereas V_{ak} specifies its contribution to electron NTO k. The source gives the illustrative combination

$$\psi_1^e = 0.75\,\phi_{LUMO} - 0.60\,\phi_{LUMO+1} + 0.15\,\phi_{LUMO+2}.$$

The electron coefficients (0.75, −0.60, 0.15) have squared norm 0.945. Their normalized version is

$$\psi_1^e = \frac{0.75\phi_{LUMO} - 0.60\phi_{LUMO+1} + 0.15\phi_{LUMO+2}}{\sqrt{0.945}}\,.$$

This first electron orbital is also a linear combination of several virtual orbitals. The central point is that **U** and **V** are obtained from the SVD of the transition matrix:

$$\mathbf{T} = \mathbf{U}\mathbf{\Lambda}\mathbf{V}^{\dagger}.$$

The matrix **U** supplies the coefficients for combinations of occupied orbitals, and **V** supplies those for combinations of virtual orbitals. This can be represented schematically as

$$original\ occupied\ orbitals\ \rightarrow [\mathrm{U}]\ hole\ NTOs$$

and

$$original\ virtual\ orbitals\ \rightarrow [\mathrm{V}]\ electron\ NTOs$$

If there are three occupied orbitals,

$$\phi_1^{occ}, \quad \phi_2^{occ}, \quad \phi_3^{occ},$$

and the first column of **U** is illustrated by the coefficient vector

$$\begin{pmatrix} 0.8 \\ 0.5 \\ -0.2 \end{pmatrix},$$

then the corresponding linear combination is

$$\psi_1^h = 0.8\phi_1^{occ} + 0.5\phi_2^{occ} - 0.2\phi_3^{occ}.$$

Thus, each column of **U** generates a hole NTO, and each column of **V** generates an electron NTO. The illustrative vector above must be normalized before it can be an actual column of a unitary matrix.

$$u_1 = \frac{1}{\sqrt{0.93}}\begin{bmatrix} 0.8 \\ 0.5 \\ -0.2 \end{bmatrix}, \quad \boldsymbol{u}_1^{\dagger}\boldsymbol{u}_1 = 1.$$

An important property follows from the unitarity of **U** and **V** (orthogonality in the real case): the NTOs remain orthonormal. Here, l is a second NTO index, $\langle f|g\rangle$ is the orbital inner product, and the Kronecker delta δ_{kl} equals 1 when $k = l$ and 0 otherwise:

$$\left\langle \psi_k^h | \psi_l^h \right\rangle = \delta_{kl}$$

and

$$\left\langle \psi_k^e | \psi_l^e \right\rangle = \delta_{kl}\,.$$

Let j be a second occupied-orbital index. Orthonormality of the original orbitals gives $\langle \phi_i^{occ} | \phi_j^{occ} \rangle = \delta_{ij}$. Substitution and contraction of this Kronecker delta give

$$\langle \psi_k^h | \psi_l^h \rangle = \sum_{ij} U_{ik}^* U_{jl} \langle \phi_i^{occ} | \phi_j^{occ} \rangle = \sum_i U_{ik}^* U_{il} = (\mathbf{U}^{\dagger}\mathbf{U})_{kl} = \delta_{kl}$$

The virtual-orbital proof is identical: replace **U** by **V** and the occupied indices by virtual indices. The last equality follows because a unitary matrix has orthonormal columns. A reduced SVD also has orthonormal retained columns. These statements require orthonormal input orbitals; a nonorthogonal atomic-orbital representation requires its overlap metric. Consequently, the transformation introduces no new information. It rotates the original orbital bases to obtain a compact representation of the electronic transition.

To summarize,

$$\psi_k^h = \ optimized\ linear\ combination\ of\ occupied\ orbitals$$

and

$$\psi_k^e = \ optimized\ linear\ combination\ of\ virtual\ orbitals$$

The pair

$$\boxed{\psi_k^h \rightarrow \psi_k^e}$$

forms the kth principal channel of the electronic transition. Now consider the transition matrix

$$\mathbf{T} = \begin{pmatrix} 0.6 & 0.3 \\ -0.4 & 0.2 \end{pmatrix}.$$

Its rows correspond to the occupied orbitals ϕ_1^{occ} and ϕ_2^{occ}, and its columns correspond to the virtual orbitals ϕ_1^{virt} and ϕ_2^{virt}. The numerical calculated singular value decomposition is

$$\mathbf{T} = \mathbf{U}\mathbf{\Lambda}\mathbf{V}^{\dagger},$$

where

$$U \approx \begin{bmatrix} 0.83 & -0.56 \\ -0.56 & -0.83 \end{bmatrix} \text{and } V \approx \begin{bmatrix} 0.98 & -0.20 \\ 0.20 & 0.98 \end{bmatrix}$$

These matrices do not constitute an SVD of the stated T with nonnegative diagonal singular values. To make the example mathematically consistent, use the following computed factors, with entries rounded to six decimal places:

$$U \approx \begin{bmatrix} 0.886114 & 0.463467 \\ -0.463467 & 0.886114 \end{bmatrix}$$

and

$$V \approx \begin{bmatrix} 0.972064 & -0.234715 \\ 0.234715 & 0.972064 \end{bmatrix}$$

$$\Lambda \approx \begin{bmatrix} 0.737663 & 0 \\ 0 & 0.325352 \end{bmatrix}$$

The corrected matrices above replace the inaccurate factors in all subsequent numerical orbital combinations. Values are rounded only for display. Simultaneously changing the sign of corresponding columns of U and V leaves their outer product unchanged.
The first column of U is

$$\begin{bmatrix} 0.886114 \\ -0.463467 \end{bmatrix}$$

It supplies the coefficients of the first hole NTO:

$$\psi_1^h \approx 0.886114\phi_1^{occ} - 0.463467\phi_2^{occ}$$

The second column of U is

$$\begin{bmatrix} 0.463467 \\ 0.886114 \end{bmatrix}$$

so

$$\psi_2^h \approx 0.463467\phi_1^{occ} + 0.886114\phi_2^{occ}$$

For the virtual orbitals, the first column of V is

$$\begin{bmatrix} 0.972064 \\ 0.234715 \end{bmatrix}$$

and therefore

$$\psi_1^e \approx 0.972064\phi_1^{virt} + 0.234715\phi_2^{virt}$$

The second column of V gives

$$\psi_2^e \approx -0.234715\phi_1^{virt} + 0.972064\phi_2^{virt}$$

The reason for using columns becomes explicit in matrix notation. Write the occupied orbitals as a row vector of functions:

$$\boldsymbol{\phi}^{\mathrm{occ}} = (\phi_1^{\mathrm{occ}} \quad \phi_2^{\mathrm{occ}}).$$

Then the hole NTOs are

$$\left(\psi_1^h \quad \psi_2^h\right) = (\phi_1^{\mathrm{occ}} \quad \phi_2^{\mathrm{occ}})\mathbf{U}.$$

Substituting the corrected matrix U gives

$$[\psi_1^h \ \psi_2^h] \approx [\phi_1^{occ} \ \phi_2^{occ}] \begin{bmatrix} 0.886114 & 0.463467 \\ -0.463467 & 0.886114 \end{bmatrix}$$

Carrying out the multiplication, the first component is

$$\psi_1^h \approx 0.886114\phi_1^{occ} - 0.463467\phi_2^{occ}$$

because it uses the first column of U. The second component is

$$\psi_2^h \approx 0.463467\phi_1^{occ} + 0.886114\phi_2^{occ}$$

because it uses the second column. This is the usual rule for matrix multiplication: each component of the resulting row vector is obtained by combining the original orbital row vector with one column of the transformation matrix. The same reasoning applies to virtual orbitals:

$$(\psi_1^e \quad \psi_2^e) = \left(\phi_1^{\mathrm{virt}} \quad \phi_2^{\mathrm{virt}}\right)\mathbf{V}.$$

Hence,

$$\psi_k^e = \sum_a V_{ak}\, \phi_a^{\mathrm{virt}}.$$

The index k appears second in U_{ik} and V_{ak} because it identifies the column. For fixed k, vary i or a:

$$\psi_k^h = U_{1k}\phi_1^{occ} + U_{2k}\phi_2^{occ} + \cdots$$

and

$$\psi_k^e = V_{1k}\phi_1^{virt} + V_{2k}\phi_2^{virt} + \cdots$$

Therefore,

$$k = \ column\ index \ = \ NTO\ index$$

There is a further interpretation. In the SVD,

$$\mathbf{T} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^{\dagger},$$

we may write

$$\mathbf{T} = \lambda_1 \mathbf{u}_1 \mathbf{v}_1^\dagger + \lambda_2 \mathbf{u}_2 \mathbf{v}_2^\dagger + \cdots$$

where $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots$ are the columns of $\mathbf{U}$, and $\mathbf{v}_1, \mathbf{v}_2, \ldots$ are the columns of $\mathbf{V}$. Thus, each term

$$\lambda_k \mathbf{u}_k \mathbf{v}_k^\dagger$$

represents a transition channel

$$\boxed{\psi_k^h \to \psi_k^e}$$

whose amplitude is characterized by λ_k, the kth singular value, and whose weight is often characterized by λ_k^2. Thus, columns are used because the SVD decomposes $\mathbf{T}$ into pairs of column vectors:

$$\boxed{\mathbf{T} = \sum_k \lambda_k \mathbf{u}_k \mathbf{v}_k^\dagger}$$

These column vectors are precisely the coefficients used to construct the NTOs. A nonzero outer product multiplied by λ_k is a rank-one contribution. Its squared Frobenius norm is λ_k^2, because both singular vectors have unit norm. For $\mathbf{T} \neq 0$, define the normalized pair weight w_k by

$$w_k = \frac{\lambda_k^2}{\sum_j \lambda_j^2}, \quad \sum_k w_k = 1, \quad \|\mathbf{T}\|_F^2 = \sum_{ia} |T_{ia}|^2 = \sum_k \lambda_k^2$$

Here, j is a dummy pair index, and $\|\mathbf{T}\|_F$ is the Frobenius norm. The normalized weights measure the share of the matrix norm carried by each pair; they are not automatically literal probabilities of electron motion. If $\mathbf{T} = 0$, normalized weights are undefined and no nonzero transition channel exists.

6 Another way to derive the NTOs

Let a ket $|f\rangle$ denotes an orbital vector, a bra $\langle f|$ its Hermitian adjoint, and $\hat{T}$ the one-electron operator associated with T. The derivation starts from the matrix entries

$$T_{ia}$$

and arrives at the NTO operator's representation

$$\hat{T} = \sum_k \lambda_k |\psi_k^h\rangle\langle\psi_k^e|$$

This derivation explains why hole-electron pairs arise. Proceeding carefully makes the construction of NTOs mathematically transparent. Begin with the transition matrix **T**, whose entries are

$$T_{ia},$$

where i runs over occupied orbitals and a runs over virtual orbitals.

A ket $|f\rangle$ represents an orbital vector, and its bra $\langle f|$ is its Hermitian adjoint. An outer product $|f\rangle\langle g|$ maps $|g\rangle$ into $|f\rangle$ when g is normalized. We write $\hat{T}$ for the one-electron operator associated with the matrix **T**, distinguishing it from the matrix itself. The following occupied-row, virtual-column operator maps the virtual space into the occupied space. Its adjoint maps occupied into virtual space. Neither should be confused with the many-electron excitation operator $\hat{a}_a^\dagger \hat{a}_i$.

$$\hat{T} = \sum_{ia} T_{ia} \left|\phi_i^{\mathrm{occ}}\right\rangle\langle\phi_a^{\mathrm{virt}}\right|.$$

Each entry T_{ia} associates occupied orbital i with virtual orbital a. Applying the SVD gives

$$\boxed{\mathbf{T} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^\dagger}$$

where the rectangular diagonal matrix has the form

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots \\ 0 & \lambda_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

In terms of matrix elements,

$$T_{ia} = \sum_k U_{ik}\, \lambda_k V_{ak}^*.$$

This step is fundamental: the collection of amplitudes T_{ia} is rewritten as a sum of contributions indexed by k. Substitute this identity into the operator expression:

$$\hat{T} = \sum_{ia} \left(\sum_k U_{ik}\, \lambda_k V_{ak}^* \right) \left|\phi_i^{\mathrm{occ}}\right\rangle\langle\phi_a^{\mathrm{virt}}\right|.$$

Because the orbital basis is finite, the sums can be reordered without convergence issues:

$$\hat{T} = \sum_k \lambda_k \sum_{ia} U_{ik}\, V_{ak}^* \left|\phi_i^{\mathrm{occ}}\right\rangle\langle\phi_a^{\mathrm{virt}}\right|.$$

Distribute the products and separate the sum over i from the sum over a:

$$\hat{T} = \sum_k \lambda_k \left(\sum_i U_{ik} \left|\phi_i^{\mathrm{occ}}\right\rangle \right) \left(\sum_a V_{ak}^* \left\langle\phi_a^{\mathrm{virt}}\right| \right).$$

Two linear combinations now emerge naturally.
Define

$$|\psi_k^h\rangle = \sum_i U_{ik}|\phi_i^{occ}\rangle$$

and

$$|\psi_k^e\rangle = \sum_a V_{ak}|\phi_a^{\text{virt}}\rangle$$

Taking the Hermitian adjoint conjugates the coefficients, so

$$\langle\psi_k^e| = \sum_a V_{ak}^*\langle\phi_a^{\text{virt}}|$$

Substituting these definitions gives

$$\hat{T} = \sum_k \lambda_k|\psi_k^h\rangle\langle\psi_k^e|$$

Indeed, applying the operator to an electron NTO makes the pairing explicit: orthonormality gives

$$\hat{T}|\psi_l^e\rangle = \sum_k \lambda_k|\psi_k^h\rangle\langle\psi_k^e|\psi_l^e\rangle = \lambda_l|\psi_l^h\rangle$$

$$\hat{T}^\dagger|\psi_l^h\rangle = \lambda_l|\psi_l^e\rangle$$

This is the NTO representation of the transition operator. It decomposes the entire transition into orthogonal orbital pairs:

$$\psi_1^h \leftrightarrow \psi_1^e,$$

$$\psi_2^h \leftrightarrow \psi_2^e,$$

$$\psi_3^h \leftrightarrow \psi_3^e.$$

and so on. The coefficient λ_k measures the amplitude associated with pair k. If

$$\lambda_1 \gg \lambda_2, \lambda_3, \ldots,$$

then the first pair describes nearly the entire transition:

$$\boxed{\psi_1^h \to \psi_1^e.}$$

"Nearly the entire transition" means that the normalized leading weight is close to 1. Let $T^{(p)}$ retain the first p singular terms, with $1 \le p \le r_T$. Orthogonality of the outer products gives

$$T^{(p)} = \sum_{k=1}^{p}\lambda_k\, u_k\, v_k^\dagger, \quad \|T - T^{(p)}\|_F^2 = \sum_{k=p+1}^{q}\lambda_k^2$$

This truncated SVD minimizes the Frobenius error among matrices of rank at most p. For example, singular values 0.90 and 0.10 give a leading weight of $0.81/0.82 \approx 98.78\%$, with squared residual 0.01 after retaining one pair. Retaining every nonzero pair is exact; truncation is an approximation. This is why NTO representations can be compact. The same decomposition is apparent directly in matrix form. Since

$$\mathbf{T} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^\dagger,$$

expansion in columns gives

$$\boxed{T = \lambda_1 \mathbf{u}_1 \mathbf{v}_1^\dagger + \lambda_2 \mathbf{u}_2 \mathbf{v}_2^\dagger + \cdots}$$

where

$$\mathbf{u}_k$$

is the kth column of U, and

$$\mathbf{v}_k$$

is the kth column of V. Each outer product

$$\mathbf{u}_k \mathbf{v}_k^\dagger$$

is a rank-one matrix: its columns are all proportional to the nonzero vector u_k, and its image is one-dimensional. Thus, the SVD expresses the following statement:

$$transition\ matrix\ =\ sum\ of\ rank-one\ transition\ contributions$$

Each such rank-one contribution corresponds to an NTO pair. There is also an interpretation in terms of transition density. Let r denote the three-dimensional position vector, and let the superscript "tr" label transition density. With the operator convention used here, the transition density is the diagonal of its position-space kernel.

The real-orbital form is unchanged. The variable r is shown as a position argument; a second position r' would give the full kernel $\langle r|\hat{T}|r'\rangle$, and setting $r' = r$ gives its diagonal.

$$\rho^{tr}(r) = \sum_{ia} T_{ia} \phi_i^{occ}(r) [\phi_a^{virt}(r)]^*$$

Substituting the matrix-element form of the SVD,

$$T_{ia} = \sum_k U_{ik}\, \lambda_k V_{ak}^*,$$

gives

$$\rho^{tr}(r) = \sum_{iak} U_{ik} \lambda_k\, V_{ak}^* \phi_i^{occ}(r) [\phi_a^{virt}(r)]^*$$

Reordering and factoring the finite sums give

$$\rho^{tr}(r) = \sum_k \lambda_k [\sum_i U_{ik} \phi_i^{occ}(r)] [\sum_a V_{ak}^* [\phi_a^{virt}(r)]^*]$$

Recognizing the NTOs and the complex-conjugated electron NTO yields

$$\boxed{\rho^{tr}(\mathbf{r}) = \sum_k \lambda_k\, \psi_k^h(\mathbf{r}) \psi_k^{e*}(\mathbf{r}).}$$

If the orbitals and coefficients are real, this becomes

$$\boxed{\rho^{tr}(\mathbf{r}) = \sum_k \lambda_k \psi_k^h(\mathbf{r})\psi_k^e(\mathbf{r}).}$$

This identity shows that NTOs do not change the transition density. They express it in another orbital basis that is often much more compact. In other words,

$$canonical\ orbitals \rightarrow T\ matrix \rightarrow SVD \rightarrow NTO\ pairs$$

and

$$\hat{T} = \sum_k \lambda_k |\psi_k^h\rangle\langle\psi_k^e|$$

summarizes the construction mathematically. Depending on the convention, the literature may reverse the order of the bra and ket, for example,

$$\left|\psi_k^e\right\rangle\langle\psi_k^h\left|\right.\right.$$

This depends on the definition of T, in particular, whether occupied orbitals label rows and virtual orbitals label columns, or vice versa. The physical content is unchanged when the entire convention is transformed consistently.

7 Added numerical reconstruction of the 2 × 2 example

For the matrix with rows (0.6, 0.3) and (−0.4, 0.2), compute the two products entry by entry. For example, the upper-right entry of $TT^\dagger$ is $0.6(-0.4) + 0.3(0.2) = -0.18$. The results are

$$\mathbf{TT}^\dagger = \begin{bmatrix} 0.45 & -0.18 \\ -0.18 & 0.20 \end{bmatrix}, \quad \mathbf{T}^\dagger\mathbf{T} = \begin{bmatrix} 0.52 & 0.10 \\ 0.10 & 0.13 \end{bmatrix}$$

Let μ denote an eigenvalue of either product and let I_2 be the 2 × 2 identity matrix. The determinant condition for a nonzero eigenvector is

$$det(\mathbf{TT}^\dagger - \mu\mathbf{I}_2) = (0.45 - \mu)(0.20 - \mu) - 0.18^2 = \mu^2 - 0.65\mu + 0.0576 = 0$$

$$\mu_{1,2} = \frac{0.65 \pm \sqrt{0.1921}}{2} \approx 0.54414607, 0.10585393$$

Taking the nonnegative square roots gives $\lambda_1 \approx 0.73766257$ and $\lambda_2 \approx 0.32535201$. For the first left eigenvector, write its components as x and y. The first row gives $(0.45 - \mu_1)x - 0.18y = 0$, so $y/x = (0.45 - \mu_1)/0.18 \approx -0.52303372$. Enforcing $x^2 + y^2 = 1$ and choosing $x > 0$ gives

$u_1 \approx (0.886114, -0.463467)^T$. Choose the orthogonal unit vector $u_2 \approx (0.463467, 0.886114)^T$, and obtain the corresponding right vectors from $\mathbf{v}_k = \mathbf{T}^\dagger \boldsymbol{u}_k / \lambda_k$. This produces the corrected $\mathbf{V}$ above. The two weighted outer products are

$$\lambda_1 \boldsymbol{u}_1 \mathbf{v}_1^\dagger \approx \begin{bmatrix} 0.63539274 & 0.15342237 \\ -0.33233183 & -0.08024508 \end{bmatrix}$$

$$\lambda_2 \boldsymbol{u}_2 v_2^\dagger \approx \begin{bmatrix} -0.03539274 & 0.14657763 \\ -0.06766817 & 0.28024508 \end{bmatrix}$$

Adding entries in matching positions reconstructs the matrix: $0.63539274 - 0.03539274 = 0.60$, $0.15342237 + 0.14657763 = 0.30$, $-0.33233183 - 0.06766817 = -0.40$, and $-0.08024508 + 0.28024508 = 0.20$. The equality is exact before rounding. The squared singular values sum to 0.65, giving normalized pair weights of approximately 83.71% and 16.29%. Thus, this particular example contains a substantial second pair.

8 Equivalent Formulation by Diagonalization

The SVD can also be understood through the square matrices $\mathbf{T}\mathbf{T}^\dagger$ and $\mathbf{T}^\dagger\mathbf{T}$:

$$\mathbf{T}\mathbf{T}^\dagger \boldsymbol{u}_k = \lambda_k^2 \boldsymbol{u}_k$$

$$\mathbf{T}^\dagger\mathbf{T}\mathbf{v}_k = \lambda_k^2 \mathbf{v}_k$$

The eigenvectors $\boldsymbol{u}_k$ generate the hole NTOs, and the eigenvectors $\mathbf{v}_k$ generate the electron NTOs. Both eigenvalue problems share the same nonzero eigenvalues λ_k^2, naturally establishing the hole-electron pairs.

Substituting $\mathbf{T} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^\dagger$ and using $\mathbf{V}^\dagger\mathbf{V} = \mathbf{I_m}$ and $\mathbf{U}^\dagger\mathbf{U} = \mathbf{I_n}$, where $\mathbf{I}_m$ and $\mathbf{I}_n$ are identity matrices of the indicated sizes, gives

$$\mathbf{T}\mathbf{T}^\dagger = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^\dagger\mathbf{V}\boldsymbol{\Lambda}^\dagger\mathbf{U}^\dagger = \mathbf{U}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\dagger)\mathbf{U}^\dagger$$

$$\mathbf{T}^\dagger\mathbf{T} = \mathbf{V}\boldsymbol{\Lambda}^\dagger\mathbf{U}^\dagger\mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^\dagger = \mathbf{V}(\boldsymbol{\Lambda}^\dagger\boldsymbol{\Lambda})\mathbf{V}^\dagger$$

The diagonal entries of $\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\dagger$ and $\boldsymbol{\Lambda}^\dagger\boldsymbol{\Lambda}$ are squared singular values, with additional zeros when the matrix is rectangular. Multiplication by the kth column gives the two eigenvalue equations above. For each $\lambda_k > 0$, pairing follows from

$$\mathbf{T}\mathbf{v}_\mathrm{k} = \lambda_\mathrm{k}\,\mathbf{u}_\mathrm{k}, \quad \mathbf{T}^\dagger\mathbf{u}_\mathrm{k} = \lambda_\mathrm{k}\,\mathbf{v}_\mathrm{k}, \quad \mathbf{v}_\mathrm{k} = \frac{\mathbf{T}^\dagger\mathbf{u}_\mathrm{k}}{\lambda_\mathrm{k}}$$

For example, the products in the preceding 2 × 2 calculation have eigenvalues 0.54414607 and 0.10585393. Their square roots reproduce its singular values. If several singular values are equal, rotations within the corresponding subspaces are possible; individual pairs are not unique, although the combined contribution is unchanged. Null-space vectors with zero singular value do not define a nonzero transition channel.

9 Relationship to the Transition Density

For real orbitals and coefficients, let $\rho_{0I}^{tr}(\boldsymbol{r})$ denote the transition density between the reference state, labeled 0, and the selected excited state, labeled I. Suppressing the "occ" and "virt" labels on the original orbitals, the expression is

$$\rho_{0I}^{tr}(\boldsymbol{r}) = \sum_{ia} T_{ia} \phi_i(\boldsymbol{r}) \phi_a(\boldsymbol{r})$$

After the SVD, the same information can be written more compactly:

$$\rho_{0I}^{\mathrm{tr}}(\mathbf{r}) = \sum_k \lambda_k \, \psi_k^h(\mathbf{r}) \psi_k^e(\mathbf{r})$$

This is the essential reason for using NTOs: they reorganize the same electronic transition into hole-electron pairs that often provide a much simpler interpretation than dozens of canonical-orbital excitations.

For a single nonzero entry $T_{ia} = c$, where c is a real positive amplitude, one NTO pair suffices and $\rho_{0I}^{tr}(\boldsymbol{r}) = c\phi_i(\boldsymbol{r})\phi_a(\boldsymbol{r})$. The spatial product can change sign and is not a probability density. Its integral is zero when the occupied and virtual orbitals are orthogonal. Transition density describes coherence between states; it is distinct from the excited-state minus ground-state density. Hole and electron NTOs indicate removal and attachment character, but their individual orbital densities do not by themselves equal the full density difference in a general correlated or response-theory treatment.

10 A Note on TD-DFT and TDA

In the Tamm–Dancoff approximation (TDA), only the excitation amplitudes X appear, and the teaching model can be written as

$$T_{ia} \approx X_{ia}$$

Full TD-DFT contains excitation amplitudes X and de-excitation amplitudes Y. The exact transition matrix used to generate NTOs can depend on the theoretical formulation and the program implementation. Therefore, do not combine all arrows or coefficients printed in the output without checking the software's convention.

In a stated TDA convention with $\mathbf{T} = \mathbf{X}$ and $\mathbf{X} = [0.8 \quad 0.6]$, the only singular value is 1. The hole NTO is the single occupied orbital, and the electron NTO is $0.8\phi_1^{\mathrm{virt}} + 0.6\phi_2^{\mathrm{virt}}$. In a full response calculation, a program might form a spin-summed transition-density quantity involving $\mathbf{X} + \mathbf{Y}$; this is not a universal instruction to add printed amplitudes. Check the state, spin adaptation, normalization, and matrix orientation in that program's documentation.

All numerical excitation examples above use real amplitudes. For complex CIS coefficients defined by $|\Psi_I\rangle = \sum_{ia} X_{ia} \hat{a}_a^\dagger \hat{a}_i |\Phi_0\rangle$, the occupied-row matrix associated with the excitation-directed

one-electron operator's adjoint uses conjugated coefficients, $T_{ia} = X_{ia}^{*}$. Alternatively, defining the opposite transition-density orientation changes which coefficients are conjugated. The algebraic SVD identities hold for any consistently defined complex **T**; the shorthand $\mathbf{T} = \mathbf{X}$ must not override that convention. Spatial spin adaptation can also introduce normalization factors.

11 Practical Summary

1. Select the relevant occupied orbitals and fix the row order.
2. Select the relevant virtual orbitals and fix the column order.
3. For each excitation $i \to a$, place its signed amplitude in entry T_{ia}.
4. For missing or negligible excitations, use zero only as a approximation.
5. Do not square the coefficients when constructing T.
6. Apply the SVD, $\mathbf{T} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^{\dagger}$.
7. Use **U** to construct the hole NTOs and **V** to construct the electron NTOs.
8. Examine λ_k^2 to assess the importance of each NTO pair.

When **T** is not normalized, use $w_k = \lambda_k^2 / \sum_j \lambda_j^2$ for relative contributions. In the corrected 2 × 2 calculation, these weights are 83.71% and 16.29%; in an exact one-pair example, the only nonzero weight is 100%.

$$excitation\ amplitudes \rightarrow T \rightarrow SVD \rightarrow NTOs$$

An entry T_{ia} is neither an orbital nor a probability. It is the amplitude associated with an excitation connecting the occupied and virtual spaces. The SVD finds new linear combinations within the two spaces that make this connection diagonal and as compact as possible. These combinations are the NTOs.

12 Supplementary reading

For a software-level description of NTO construction and interpretation:

1 Q-Chem 7.0 User's Manual, Section 7.13.3, "Natural Transition Orbitals," *https://manual.q-chem.com/latest/sect_NTOs.html*.

2 For formal reduced-density-operator foundations: Thibaud Etienne, "Natural-orbital representation of molecular electronic transitions," arXiv:2104.11947v2 (2022), https://arxiv.org/abs/2104.11947.